ComicCraft

AI Comic Story Creator using Gemini Models

Project Documentation

An AI-powered web application that transforms story ideas into structured comic books with panel-by-panel narration, dialogue, AI-generated artwork, live preview, and A4 PDF export.

Technology Stack

FastAPI • Google Gemini • Stable Diffusion / Diffusers • FPDF • HTML Templates • Python

Prepared as an A4 project report

1. Project Overview

ComicCraft is an AI-powered web application that automatically transforms story ideas into fully structured comic books. It generates a comic outline, expands the story into narration and dialogue, creates matching comic-style illustrations, combines the content into a layout, and exports the final comic as a multi-page PDF.

Primary technologies: FastAPI backend, Google's Gemini models, Stable Diffusion through Diffusers, and FPDF.

2. Technical Architecture & Core Workflow

Step	Component	Purpose
1	Gemini Flash	Generates a structured 5-panel comic outline in JSON.
2	Gemini Pro	Expands the outline into full narration and character dialogue.
3	Diffusers / Stable Diffusion	Generates comic-style illustrations for each panel.
4	Layout Builder	Combines panel images with narrative text.
5	FPDF Exporter	Creates the final multi-page PDF document.

Core workflow:

> [User Form / JSON API] → [FastAPI Backend] → [Gemini Flash: 5-Panel Outline] → [Gemini Pro: Story & Dialogues] → [Stable Diffusion: Artwork] → [Layout Builder] → [FPDF Exporter] → [A4 Comic PDF]

3. Key Scenarios

Scenario 1 – Custom Story Generation

A user submits a custom prompt containing character details, setting, tone, and visual style. The system generates a multi-panel comic strip matching those specifications.

Scenario 2 – Iterative Style Customization

A user changes the story tone, such as dramatic to funny, and changes the art style, such as classic comic book. The generation pipeline runs again to create the redesigned narrative and visuals.

Scenario 3 – PDF Export & Confirmation

After viewing the live preview, the user downloads the comic. The application compiles panel titles, images, and text into a structured PDF using FPDF and displays a download-success page.

4. Project Structure & File Index

File / Folder	Purpose	Status
app/main.py	FastAPI application initialization	Completed
app/routes.py	API endpoints and web form route handling	Completed
app/gemini_flash.py	Structured 5-panel comic outline generation	Completed
app/gemini_pro.py	Full story narration and character dialogue generation	Completed
app/image_generator.py	Stable Diffusion illustration pipeline	Completed
app/layout_builder.py	Combines text segments and panel images	Completed

File / Folder	Purpose	Status
app/exporters.py	Multi-page PDF compilation using FPDF	Completed
templates/index.html	User input form interface	Completed
templates/comic_preview.html	Panel-by-panel comic preview page	Completed
templates/export_success.html	Download confirmation page	Completed

5. Story Documentation: "The Secret of the Magical Library"

Topic & Character Roster

Character	Description
Arjun (15)	Curious and brave boy; group leader.
Maya (14)	Intelligent girl; loves books and solving puzzles.
Kavin (15)	Funny friend; easily scared but very loyal.
Diya (13)	Creative girl; skilled at drawing and observing hidden clues.
Rohan (16)	Strong and protective friend.
Professor Anand	Old teacher who knows the secret history of the library.
Luna	Mysterious magical girl residing in the Crystal Forest.
Eli	Small talking owl guiding the group through Elaria.
The Shadow Keeper	Mysterious guardian of the forbidden section.
King Orion	Legendary spirit from the magical book world.

Story Chapters

Chapter 1 – The Old Library

In a small town stood an ancient, vine-covered library that nobody visited anymore. One rainy afternoon, Arjun, Maya, Kavin, Diya, and Rohan stayed after school for a history project.

> "Professor Anand: Some books are meant to be read. Some books are meant to be discovered."

Intrigued, the five friends visited the old building that evening.

Chapter 2 – The Hidden Door

Inside the dusty library, Maya discovered a golden symbol carved into the wall behind an old bookshelf. When Diya touched it, the bookshelf shifted to reveal a hidden underground staircase.

> "Kavin: Why do secret staircases always look scary?"

Descending the steps, they arrived before a massive golden door.

Chapter 3 – The Book That Spoke

On a nearby table lay an enormous book covered in strange symbols. As Maya opened it, the pages glowed bright light.

> "Book: Five hearts must enter. Five hearts must return. Only together can the lost kingdom be saved."

The room trembled, and a blinding light pulled all five friends into the pages.

Chapter 4 – The Magical World

They materialized in Elaria, a realm of floating islands, crystal rivers, and sky castles. A small talking owl named Eli greeted them and explained that Elaria's magical heart had vanished long ago, causing the realm's magic to fade.

> "Kavin: THE OWL JUST TALKED!"

Chapter 5 – Meeting Luna

Near the Crystal Forest, the group met Luna, who revealed that King Orion had hidden the magical heart. She provided an ancient map showing three trial locations: the Crystal Forest, the Mountain of Echoes, and the Shadow Castle.

> "Luna: You must find three magical keys before you can enter the Shadow Castle."

Chapter 6 – The Crystal Forest

The group encountered a puzzle involving five crystal stones displaying unique symbols. Combining Maya's logic and Diya's keen eye, they arranged the stones in correct sequence, unlocking a bridge and retrieving the First Magical Key.

Chapter 7 – The Mountain of Echoes

On the Mountain of Echoes, every sound echoed back as a riddle. Rohan realized the echoes were providing navigation clues. When falling rocks blocked their path, Rohan and Arjun cleared a route while Kavin kept morale high. At the summit, they secured the Second Magical Key.

Chapter 8 – The Shadow Castle

At the Shadow Castle, the mysterious Shadow Keeper challenged them with a question:

> "Shadow Keeper: What is stronger than magic?"
> "Arjun: Friendship. Because magic can disappear, but people who trust each other can keep going."

Impressed by the answer, the guardian granted them entry and the Third Magical Key.

Chapter 9 – King Orion's Secret

Inside, the spirit of King Orion revealed he had protected the magical heart from selfish misuse.

> "King Orion: The greatest power is not controlling magic. It is knowing when to use it responsibly."

Rather than entrusting the heart to one individual, the group pledged to protect it together.

Chapter 10 – Saving Elaria

Inserting the three keys into the ancient crystal door, a brilliant light washed over Elaria. Trees bloomed, rivers sparkled, and floating islands restored their alignment.

> "Luna: You didn't save Elaria because you were the strongest. You saved it because you worked together."

Chapter 11 – Returning Home

A portal opened, returning the group to the old library where only a few minutes had passed in real time.

> "Professor Anand: Some stories are not written in books. They are lived by those brave enough to enter them."

6. Core Technical Implementation

6.1 Panel Outline Generation – gemini_flash.py

```
def generate_outline(user_prompt: str) -> list:
    """
    Generates a 5-panel comic layout based on the user's story idea using Gemini.
    """
    prompt = f"""
    You are a professional AI comic planner.
    Generate a strictly formatted JSON array containing 5 panel descriptions based on:
    STORY: "{user_prompt}"

    Each JSON object must include:
    - "panel": (integer)
    - "title": (string)
    - "scene_description": (string)
    - "image_prompt": (string)
    """
    response = model.generate_content(prompt)
    output_text = response.text.strip().replace("```json", "").replace("```", "")
    return json.loads(output_text)
```

6.2 Story & Dialogue Expansion – gemini_pro.py

```
def generate_story(outline: list) -> str:
    """
    Generates detailed comic story with narration and character dialogue using Gemini 1.5 Pro.
    """
    formatted_outline = "\n".join(
        [f"{i+1}. {item}" for i, item in enumerate(outline)]
    )
    prompt = f"""
    You're a comic book writer. Write a comic-style story with engaging narration
    and character dialogues based on this outline:

    {formatted_outline}
    """
    response = model.generate_content(prompt)
    return response.text
```

6.3 Image Generation Pipeline – image_generator.py

```
def generate_image(prompt: str, filename: str = None) -> str:
    """
    Generates comic-style illustrations using Stable Diffusion.
    """
    if not filename:
        filename = sanitize_filename(prompt)

    image = pipe(prompt).images[0]
    path = f"static/panels/{filename}"
    os.makedirs(os.path.dirname(path), exist_ok=True)
    image.save(path)
    return path
```

6.4 PDF Compilation Routine – exporters.py

```
def save_pdf(layout: list) -> str:
    """
    Compiles full comic panels and story into an exportable multi-page PDF
    document using FPDF.
    """
    pdf = PDF()
    pdf.set_auto_page_break(auto=True, margin=15)

    for panel in layout:
        pdf.add_page()
        pdf.set_font('DejaVu', '', 14)
        pdf.cell(0, 10, f"Panel {panel['panel']}", ln=True, align="C")

        if os.path.exists(panel['image_path']):
            pdf.image(
                panel['image_path'],
                x=10, y=30,
                w=pdf.w - 20, h=100
            )
```

```
        pdf.set_y(145)
        pdf.set_font('DejaVu', '', 12)
        pdf.multi_cell(0, 10, panel['text'])

    timestamp = datetime.now().strftime("%Y%m%d%H%M%S")
    pdf_path = os.path.join(
        EXPORT_FOLDER, f"comic_{timestamp}.pdf"
    )
    pdf.output(pdf_path)
    return pdf_path
```

7. End-to-End Process

1. The user enters a story prompt and preferences through the web interface.
2. FastAPI receives the request and sends the story idea to Gemini Flash.
3. Gemini Flash returns a structured five-panel JSON outline.
4. Gemini Pro expands the outline into narration and character dialogue.
5. Stable Diffusion generates an illustration for each panel.
6. The layout builder associates each generated image with its panel text.
7. The preview page displays the completed comic.
8. FPDF compiles the panels into a multi-page A4-compatible PDF.
9. The application returns the generated PDF through the download-success flow.

8. Conclusion

ComicCraft demonstrates how generative AI can be combined with a web backend and document-generation tools to automate the comic-creation workflow. The project integrates structured text generation, creative writing, image synthesis, web presentation, and PDF export into one pipeline.

Project Title: ComicCraft – AI Comic Story Creator using Gemini Models

Output Format: A4 PDF


ComicCraft
The Secret of the Magical Library
An AI Comic Story Creator using Gemini Models)
1. The Old Library
LIBRARY
Some books are meant to be read. Some books are meant to be discovered.
Arjun
Maya
Kavin
Diya
Rohan
Professor Anand
Five friends – Arjun, Maya, Kavin, Diya and Rohan – along with their teacher, Professor Anand, visit the old library to complete a project. What they find changes their lives forever...
Guys, look at this! This symbol isn't part of the bookshelf.
Why do secret staircases always look scary?
Maya
Kavin
Behind the bookshelf, they find a hidden staircase leading underground. At the bottom, a huge golden door awaits them.
3. The Magical World
Welcome to Elaria! The magical heart is lost, and we need your help.
Eli
Luna
The OWL just talked!
They meet Luna and Eli, who tell them that the kingdom is losing its magic. They must find three magical keys to reach the Shadow Castle.
4. The Three Keys
Crystal Forest
Mountain of Echoes
Shadow Castle
The friends must find three magical keys to reach the Shadow Castle and save Elaria.
5. The Challenges
Crystal Forest
They solve the crystal puzzle and get the first key.
Mountain of Echoes
Rohan notices the echoes give clues. They get the second key.
Shadow Castle
The Shadow Keeper asks, "What is stronger than magic?" Arjun answers: "Friendship."
7. King Orion's Secret
The greatest power is not controlling magic. It is knowing when to use it responsibly.
The friends find the magical heart and decide to protect it together.
8. Saving Elaria
You saved Elaria because you worked together.
So... does this mean we are heroes now?
They place the three keys into the crystal door. The kingdom is restored! Trees, rivers and floating islands come back to life.
10. The Lesson
Courage begins when we face the unknown, but true strength comes from friendship, teamwork and using knowledge responsibly.
The End

1. Python – Story Generator (using Gemini API)

comic_story_generator.py

```python
import google.generativeai as genai
import os

# Configure Gemini API
genai.configure(api_key="YOUR_GEMINI_API_KEY")  # Replace with your key

model = genai.GenerativeModel("gemini-1.5-pro")

def generate_story(prompt):
    response = model.generate_content(
        f"Create a detailed comic story with characters, scenes and dialogues"
        f" based on this topic: {prompt}. Write in a simple and creative style."
    )
    return response.text

# Example usage
topic = "The Secret of the Magical Library"
story = generate_story(topic)

with open("comic_story.txt", "w", encoding="utf-8") as file:
    file.write(story)

print("Story generated and saved as comic_story.txt")
```